

JEE MAIN REVISION MASTER: P-BLOCK ELEMENTS

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 25 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	P Block Elements — Core Principles & Governing Laws	14 PYQs	Core Principle
High	P Block Elements — Applied Cases, Reactions & Traps	11 PYQs	Applied Topics
Standard	P Block Elements — Standard Calculations & Problem Archetypes	1 PYQs	Quantitative Analysis

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
P Block Elements — Core Principles & Governing Laws	$\text{SbH}_3, \text{KMnO}_4, \text{NH}_3 > \text{PH}_3 > \text{AsH}_3 > \text{SbH}_3 > \text{BiH}_3$
P Block Elements — Applied Cases, Reactions & Traps	$\text{CaH}_2, \text{NH}_4\text{NO}_2, (\text{NH}_4)_2\text{Cr}_2\text{O}_7$
P Block Elements — Standard Calculations & Problem Archetypes	$\text{CaH}_2, \text{Ca}^{2+}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 P Block Elements — Core Principles & Governing Laws (High Yield — 14 PYQs)

Core Principle: Fundamental definitions, core postulates, and governing physical/chemical relations in P Block Elements.

■ **Caution / JEE Pitfall:** Carefully verify units, standard reference states, and sign conventions in P Block Elements.

■ **Speed Trick:** Focus on high-yield formulas and standard limiting approximations tested in P Block Elements PYQs.

Representative Source PYQs:

- [2024 Shift] Q#1 (MCQ): "Which of the following halides cannot be hydrolysed?..." [Ans: N/A]
- [2024 Shift] Q#3 (MCQ): "A metalloid is..." [Ans: N/A]
- [2023 Shift] Q#2 (MCQ): "The correct decreasing order of basicity of hydrides of group- 15 elements is..." [Ans: N/A]

3.2 P Block Elements — Applied Cases, Reactions & Traps (High Yield — 11 PYQs)

Core Principle: Special cases, experimental observations, and conceptual traps designed to test deep understanding of P Block Elements.

■ **Caution / JEE Pitfall:** Read questions carefully for keywords like 'incorrect', 'not true', or non-standard conditions.

■ **Speed Trick:** Elimination of extreme or unphysical answer choices improves accuracy in assertion-reason and multiple-statement questions.

Representative Source PYQs:

- [2024 Shift] Q#2 (MCQ): " NO_2 gas is..." [Ans: N/A]
- [2023 Shift] Q#1 (MCQ): "Which of the following compound does not give dinitrogen on heating?" [Ans: N/A]
- [2022 Shift] Q#1 (MCQ): " $(\text{NH}_4)_2\text{Cr}_2\text{O}_7$ on heating liberates a gas. The same gas will be obtained by..." [Ans: N/A]

3.3 P Block Elements — Standard Calculations & Problem Archetypes (Standard Yield — 1 PYQs)

Core Principle: Quantitative numerical problem-solving techniques and standard question patterns frequently seen in P Block Elements.

■ **Caution / JEE Pitfall:** Watch for algebraic rounding, stoichiometric balancing, and boundary condition checks.

■ **Speed Trick:** Dimensional consistency checks and ratio comparisons can quickly eliminate incorrect options.

Representative Source PYQs:

- [2022 Shift] Q#4 (MCQ): "Which of the following hydrides is electron deficient?" [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2024 (Q#2)

NO_2 gas is

- (A) colourless, neutral
- (B) colourless, acidic
- (C) brown, acidic
- (D) brown neutral

Correct Answer: None

Solution Analysis: NO_2 gas is brown and acidic, so the correct choice is: Option C : brown, acidic NO_2 stands for nitrogen dioxide, which is a reddish-brown gas known for its characteristic pungent smell. It is one of the primary pollutants resulting from the combustion of fossil fuels and is a contributor to atmospheric reactions that produce smog and acid rain. As an acidic gas, NO_2 can dissolve in water to form nitric and nitrous acids, contributing to environmental issues such as acid rain.

Benchmark Problem 2: KCET 2024 (Q#3)

A metalloid is

- (A) Bi
- (B) Sb
- (C) P
- (D) Se

Correct Answer: None

Solution Analysis: A metalloid can be defined as chemical elements whose physical and chemical properties lies in between metal and non-metal. Among the given option only antimony (Sb) is metalloid.

Benchmark Problem 3: KCET 2022 (Q#4)

Which of the following hydrides is electron deficient?

- (A) CaH_2
- (B) CH_4
- (C) B_2H_6
- (D) NaH

Correct Answer: None

Solution Analysis: An electron-deficient hydride is a compound that has fewer valence electrons than are required to completely satisfy the traditional bonding requirements based on the concept of two electrons for every single bond between atoms. To determine if a hydride is electron deficient, we must consider the valence electrons and bonding in each given molecule. Let's examine each option: Option A: CaH_2 Calcium hydride consists of calcium ions (Ca^{2+}) and hydride ions (H^-). Each calcium atom provides two electrons to form two ionic bonds with two hydride ions. This compound is not electr